

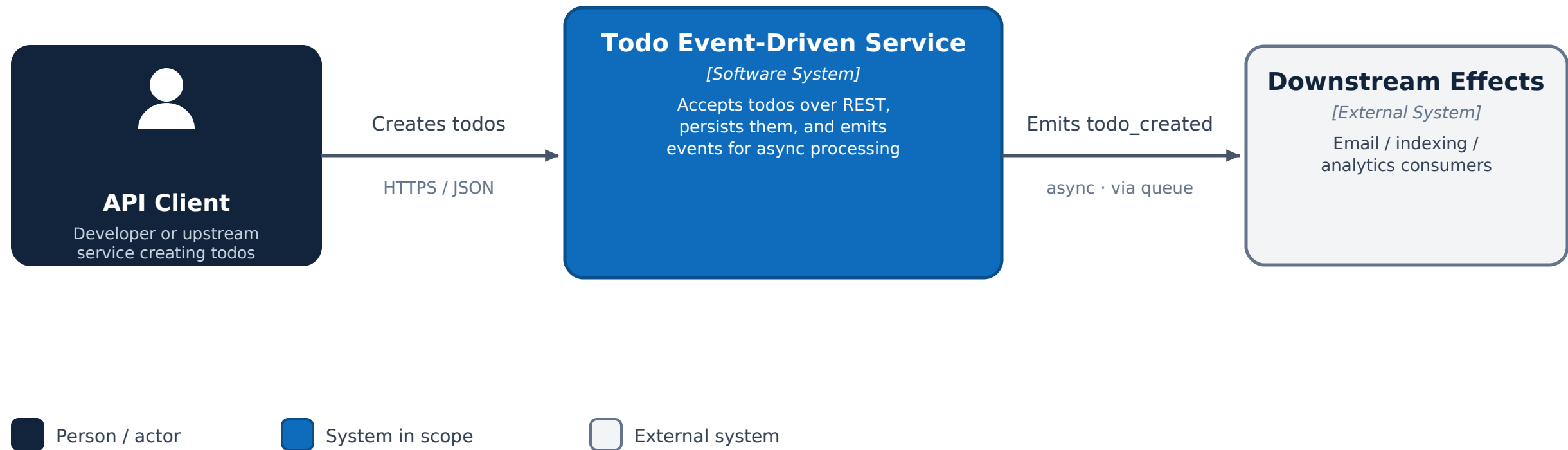
TECHNICAL ARCHITECTURE · MODULE 2

Todo Event-Driven Service

FastAPI + Azure Service Bus — System Architecture & Data-Flow

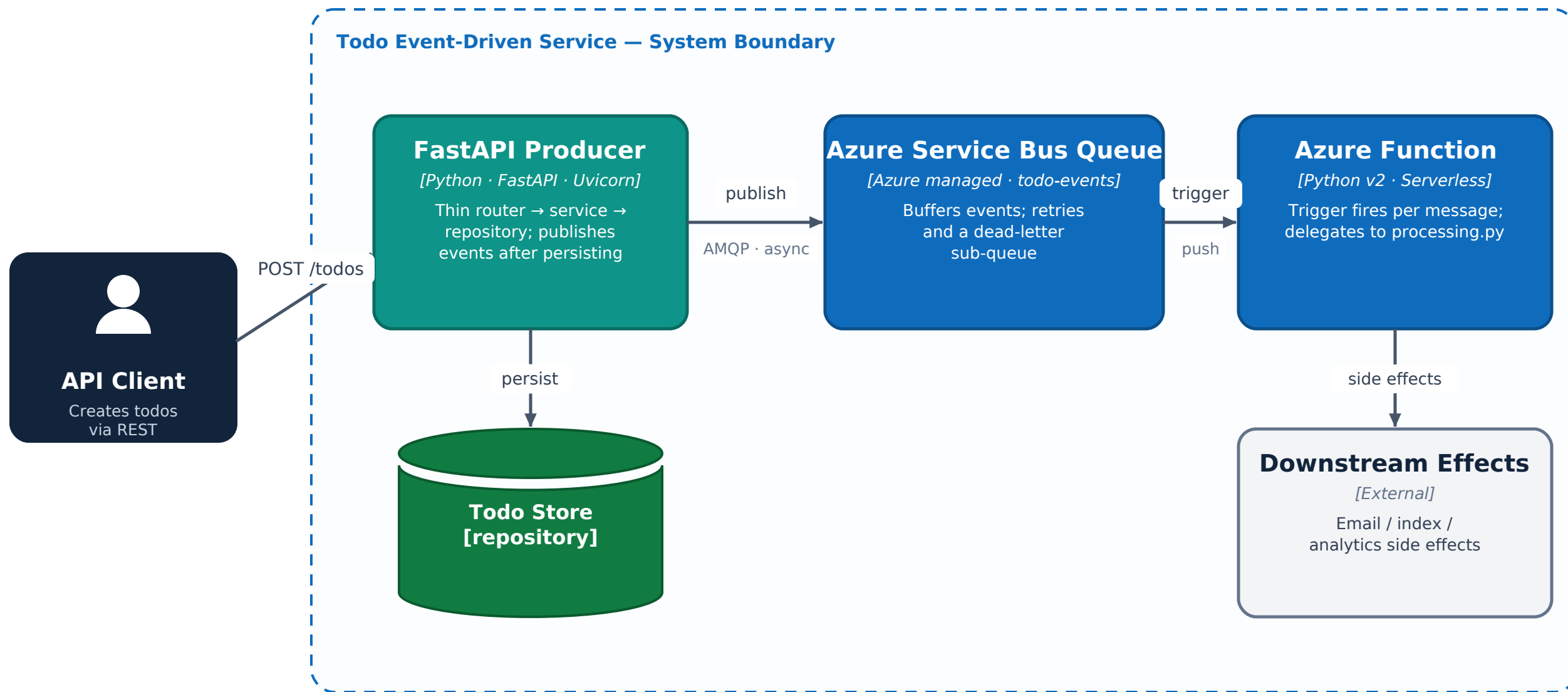
- 1 · System Context (C4 L1)
- 2 · Container Diagram (C4 L2)
- 3 · Sequence / Data-Flow (runtime)
- 4 · Cloud Deployment / Infrastructure

1 • System Context



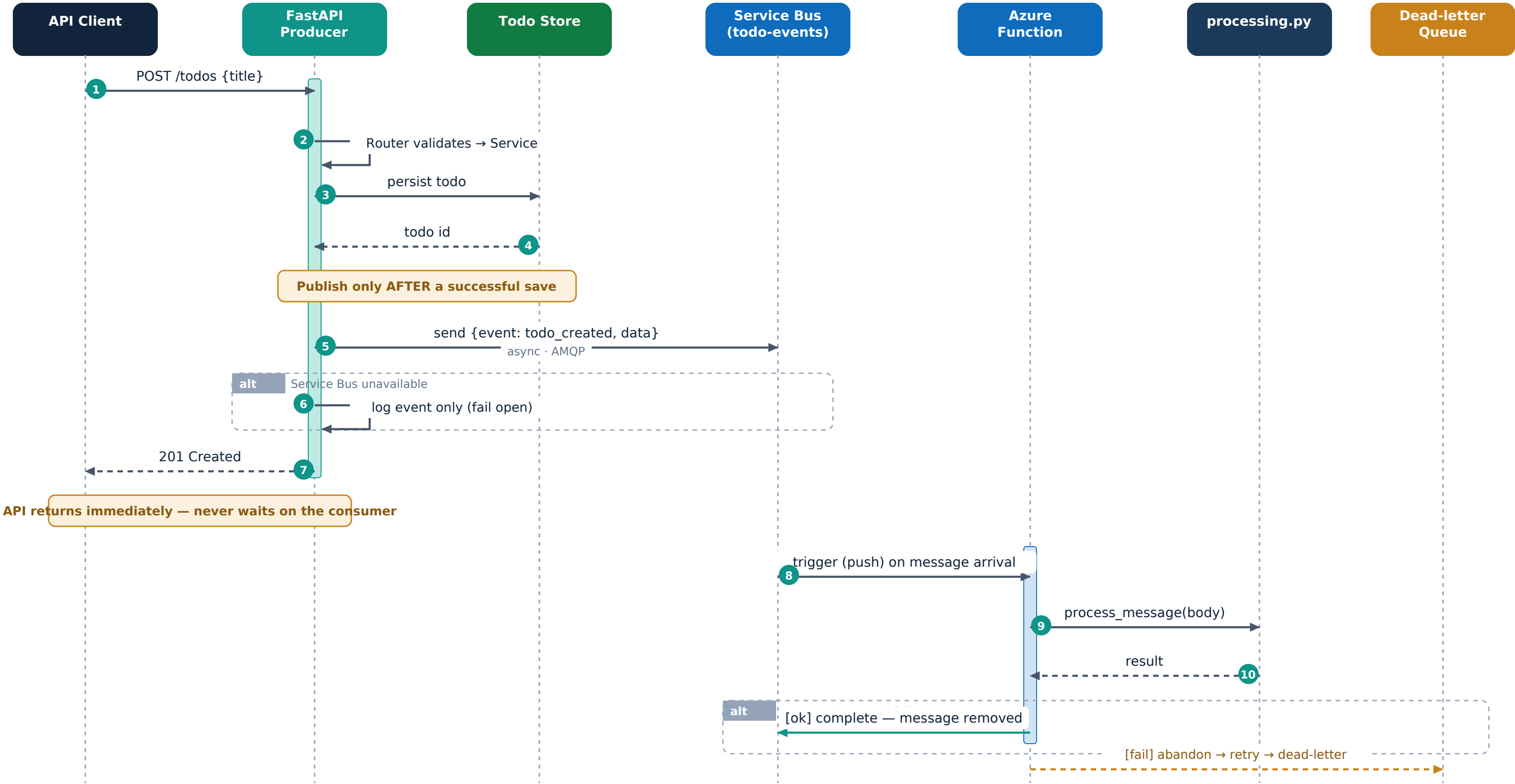
Level 1 hides the queue and consumer on purpose — at context scope, the only architectural promise is that downstream effects happen asynchronously.

2 · Container Diagram



Publish happens only AFTER the store write. The queue is a first-class container between producer and a serverless consumer that scales to zero when idle.

3 · Sequence / Data-Flow



Four encoded decisions: persist before publish · async non-blocking publish · 201 returns without waiting · platform handles retry + dead-letter.

4 · Cloud Deployment / Infrastructure Mapping

